

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	I	catalyst in the same physical state as reaction	1			1		
			II	award (1) each for any two of following - transition metals have empty orbitals so can form bonds to reactant molecules / form complex - transition metals have variable oxidation states (so they can oxidise/reduce the reactants) - products are released and the metal returns to the original oxidation state	2			2		
		(ii)		$[H^+]^2 = K_a \times [\text{acid}]$ $[H^+] = 1.97 \times 10^{-3} \text{ mol dm}^{-3}$ $\text{pH} = 2.7$		3		3	3	
				Question 10 total	8	4	3	15	6	4